

### \subsection{Approaches in Classification-Based NTL Detection}

Classification-based NTL detection utilizes a comparative approach between forecasted and actual energy consumption values, where a significant discrepancy often indicates potential fraud \cite{messinis2018unsupervised}. Both ARIMA and ARMA models, renowned for time series forecasting, play a pivotal role in this process. Badrinath et al., \cite{badrinath2016arima} found that ARIMA tends to be more effective for residential consumers. Expanding on this, Krishna et al., \cite{krishna2016f}, investigated the application of Kullback-Leibler Divergence (KLD) for detecting sophisticated energy theft, where KLD is used to assess the disparity between a set of measurements and a historical baseline. This method is particularly adept at uncovering smart attacks that blend anomalous usage into normal patterns using legitimate ARIMA models. Jindal et al. \cite{jindal2016decision} explored decision trees and SVM-based classifiers for in-depth analysis of energy consumption data to pinpoint fraud. Their approach integrates two levels of data processing, with the decision tree output serving as input for the SVM classifier. Villar-Rodriguez et al., \cite{villar2017detection} developed a novel algorithm for outlier detection in SGs, which accommodates irregularities in consumer consumption habits by focusing on consumption trends rather than temporal attributes.

However, protecting consumer privacy while detecting energy theft is a critical aspect of SGs, as highlighted by Mirzaee et al. \cite{mirzaee2022smart} and Llaría et al. \cite{llaria2021intelligent}. They proposed a Lower-Upper Decomposition (LUD) algorithm to calculate consumer honesty coefficients in a privacy-preserving manner. The LUD approach is unique in that it gauges the likelihood of having multiple energy thieves in a community based on the honesty coefficient vector. Despite its innovative approach, the LUD algorithm has limitations, such as not accounting for technical losses and being constrained by the dimensionality of consumption data, which might necessitate adjustments in data granularity.

Moreover, some classification-based methods are susceptible to contamination attacks, where energy thieves subtly alter data to mislead learning algorithms. The dependency of machine learning-based methods on extensive long-term monitoring often results in a delay in theft detection due to the requirement of large sample sizes \cite{jokar2015electricity}. Additionally, most NTL detection schemes overlook technical losses, which could impede their practical application.

To overcome these limitations, this thesis proposes an electricity theft detection (ETD) framework for smart homes with knowledge-based synthetic attack classifiers and evaluated with real attack data. These frameworks are capable of identifying electricity theft and meter irregularities without being constrained by the dimensions or granularity of power consumption data. They are designed to detect anomalies in the appliance consumption patterns, an NTL, occurring minutely or intermittently throughout the day. The proposed frameworks also consider the impact of electricity theft attacks (ETA) at three different homes assumed to have different appliance configurations and different ETA scenarios, aiming for a high detection rate with minimal false positives. A range of NTL attack scenarios

is evaluated to validate the reliability of these frameworks in real-world AMI fraud and metering defect scenarios. Importantly, these frameworks can be implemented without additional hardware costs, offering a cost-effective solution for UPs and ultimately benefiting consumers through reduced operational costs.

This chapter discusses an overview of the Smart Grid, the Advanced Metering Infrastructure (AMI) and different types of attacks. We will discuss Non-Intrusive Appliance Load Monitoring

(NIALM) - Disaggregated Smart Meter and consumption patterns of some appliances and also mention the machine learning models' performance evaluation matrix.

## \section{Advances in Non-Intrusive Load Monitoring}

In the realm of Smart Grids, AMI plays a pivotal role in generating data at the household level concerning power consumption. However, appliance-level data, offering more granularity, are increasingly sought after by homeowners and building managers for a detailed analysis of each appliance's contribution to total energy usage and cost. The conventional method of collecting such data is known as intrusive load monitoring, involving the physical installation of sensors on each appliance. While this approach yields precise usage data for each appliance, it is limited by its lack of scalability, efficiency, and cost-effectiveness.

Contrastingly, Non-Intrusive Load Monitoring (NILM) employs software algorithms to disaggregate whole-house data, effectively deducing appliance-level power consumption. This innovative technique was first introduced in the literature \cite{zoha2012non}, where it focuses on identifying changes in voltage and current to ascertain when different appliances are turned on and off. An illustrative example Figure 2.6 in Hart et al., \cite{kipreos2011nonintrusive} demonstrates how NILM algorithms discern individual appliance events based on their impact on overall household energy consumption.

NILM algorithms can be categorized into three distinct types based on their objectives:

### \begin{itemize}

\item \textbf{On/Off Classification:} This model simplifies the task of determining whether an appliance is on or off, disregarding variations during operation. Its strengths lie in high accuracy and simplicity. Due to the binary states of each appliance (on or off), these states are easier to detect, reducing the complexity of the models. However, this approach might lead to inaccurate estimations of energy consumption by treating all active states uniformly.

\item \textbf{Multiple States Recognition:} Often associated with Markov Chain models as seen in Liu et al.\cite{liu2016complex}, and Azaza et al. \cite{azaza2017evaluation}, this approach, akin to a Finite State Machine (FSM), differentiates various operational states of an appliance. It aims to identify the specific state an appliance is in during operation, thus capturing more nuanced activities. However, the challenge lies in manually defining the power consumption ranges for each state.

\item \textbf{Wave Reconstruction:} This method aspires to reconstruct the power wave of each appliance. Perfect reconstruction is challenging due to the inherent variability in appliance operation and circuit-induced noise. Furthermore, the high correlation among appliance operations can lead to compounded errors from any noise or inaccuracies.

\end{itemize}

In practical applications, the first NILM product was introduced by Enetecs, Inc., a certified meter Data service provider, with the launch of their SPEED software in 1996 ([www.enetecs.com](http://www.enetecs.com)). Since then, numerous energy service companies have offered NILM services globally, catering to both residential and commercial sectors. These services encompass various applications such as fault detection in appliances, scheduling, and consumer education, providing disaggregated data in real-time, or at various intervals like hourly or minutely. For instance, the ‘Trickl’ mobile application by London Hydro Inc., a Canadian energy provider, shares hourly NILM data with its customers. According to a study by Home Energy Analysis, Inc., NILM technologies have contributed to an average energy consumption reduction of 12.8\% ([www.nilmeu.eu](http://www.nilmeu.eu)).

## \section{Evaluating Machine Learning} \subsection{Model Performance Metrics}

In machine learning, especially in the context of network attack detection, it is critical to measure performance accurately. Performance is commonly evaluated using metrics such as the Area Under The Curve (AUC) of Receiver Operating Characteristics (ROC) curves and the F1 score [3]. The F1 score is particularly crucial in scenarios where datasets are imbalanced. Relying solely on accuracy for performance evaluation in such cases is insufficient. The F1 score, nearing 1.0, signifies a robust model, representing a harmonized average of precision and recall. Additionally, the assessment includes the False Positive Rate (FPR) and False Negative Rate (FNR), which are vital for gauging the efficacy of detection systems.

## \section{Chapter Summary and Overview}

Utility Providers (UPs) globally strive to reduce operational expenses and enhance revenue, often confronting challenges like technical losses (TLs) and non-technical losses (NTLs), with energy theft being

a primary factor in NTLs. This type of loss not only inflates operational costs but also indirectly burdens honest consumers through higher energy prices. The advent of Smart Grid (SG) technologies has ushered in more effective strategies for detecting and analyzing potential energy fraud. The fine-grained data provided by Smart Meters (SMs) and other intelligent devices can be utilized through data analytics and software applications to pinpoint instances of energy fraud and metering defects with precision. The integration of these smart devices into revenue protection systems offers UPs substantial returns on their SG investments.

The current paradigm in combating energy theft involves leveraging consumers' detailed energy usage data coupled with sophisticated data analytics. This approach is akin to fraud detection mechanisms used in other industries, such as banking and credit card transactions, and is highly recommended for adoption by UPs.

This chapter has delved into the fundamentals of SGs and various aspects of Advanced Metering Infrastructure (AMI). It provided a comprehensive review of the literature concerning electricity losses in electrical distribution systems, encompassing both TL and NTL activities. It also discussed different methods of energy theft, including direct connections, meter tampering, and methods to slow down the meter's rotating disk in both low voltage (LV) and high voltage (HV) energy meters. Finally, the chapter presented a detailed examination of existing NTL detection schemes, categorizing them into state-based, game theory-based, and classification-based detection methods. Additionally, the chapter covered the advances in Non-Intrusive Load Monitoring (NILM) as a significant development in AMI, highlighting its role in enhancing the granularity of energy consumption analysis and its importance in modern SGs.